

Mitigating Catastrophic Forgetting in EN-DE Idiom Translation via Two-Stage Fine-Tuning and Metric-Aware Evaluation

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Idioms are challenging in Machine Translation

- Idioms are semantically tricky
- Literal translations will usually not convey the correct meaning

EN Idiom Sentence	DE Semantic Meaning	Literal Back Translation
I couldn't care less	Es ist mir Wurst	That's me sausage
I think I'm crazy	Ich glaube ich spinne	I think I spider
I smell a rat	Da ist etwas faul	Something is foul



Idea: Two-Stage Idiom-aware Fine-tuning

Intuition

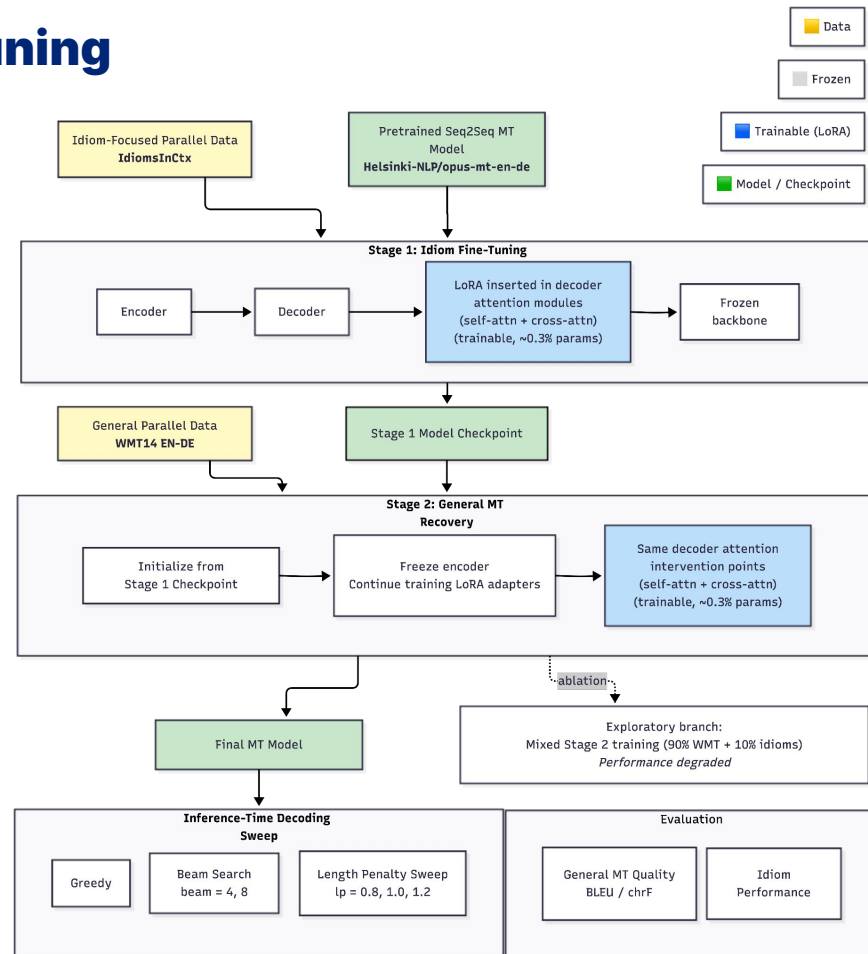
- If we fine-tune a model only on EN-DE idioms, the model will perform great on idiom translation, BUT will sacrifice general language capabilities

Approach

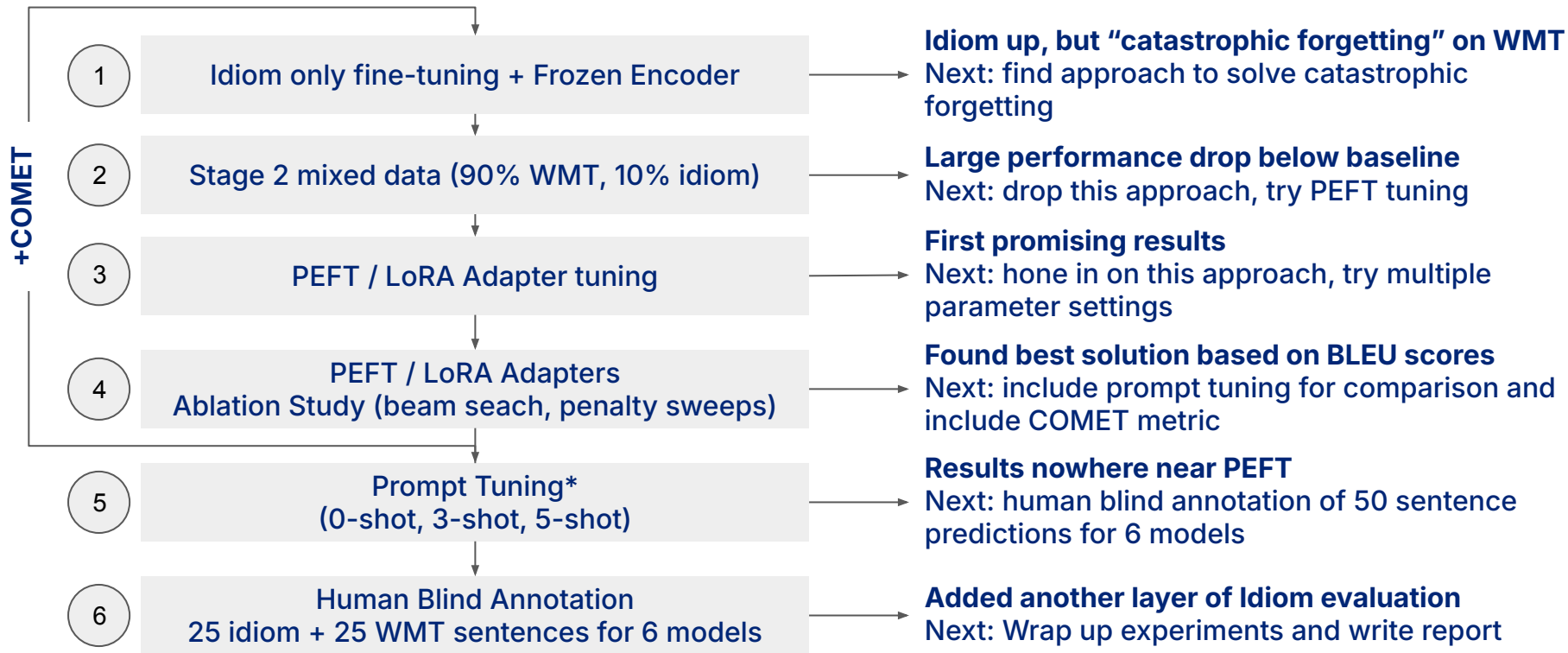
- First, fine-tune a base model on EN-DE idiom sentence pairs (IdiomsInCtx) → Stage 1
- Then, continue fine-tune the stage 1 checkpoint on EN-DE general language sentence pairs (WMT14 EN-DE) → Stage 2

Goal

- Improve idiom translation performance
- Recover lost general language capabilities without sacrificing idiom performance gains



Method and Experiments



Results

Automatic Evaluation Metrics

Model	Idioms			WMT		
	BLEU	chrF	COMET	BLEU	chrF	COMET
Baseline	39.65	60.79	0.733	27.57	58.43	0.833
Idiom-only	44.16	64.44	0.763	26.54	58.23	0.863
Two-stage	43.16	63.34	0.761	27.18	58.34	0.847
Mixed-data	43.49	62.92	0.741	26.92	58.10	0.852
LoRA-4	41.85	61.84	0.729	27.57	58.49	0.833
LoRA-8	42.18	61.76	0.732	27.48	58.46	0.842
LoRA-16	41.72	61.45	0.733	27.60	58.46	0.847
Prompt (0-shot)	8.13	33.14	0.381	7.62	35.54	0.474

Table 1: Automatic evaluation by split.

- Idiom-only fine-tuning achieves the strongest performance on idioms, while two-stage fine-tuning provides a strong balance between specialization and generalization.
- Notably, metric-dependent differences emerge: idiom-only fine-tuning decreases BLEU on WMT but improves COMET, highlighting divergence between lexical and semantic evaluation.

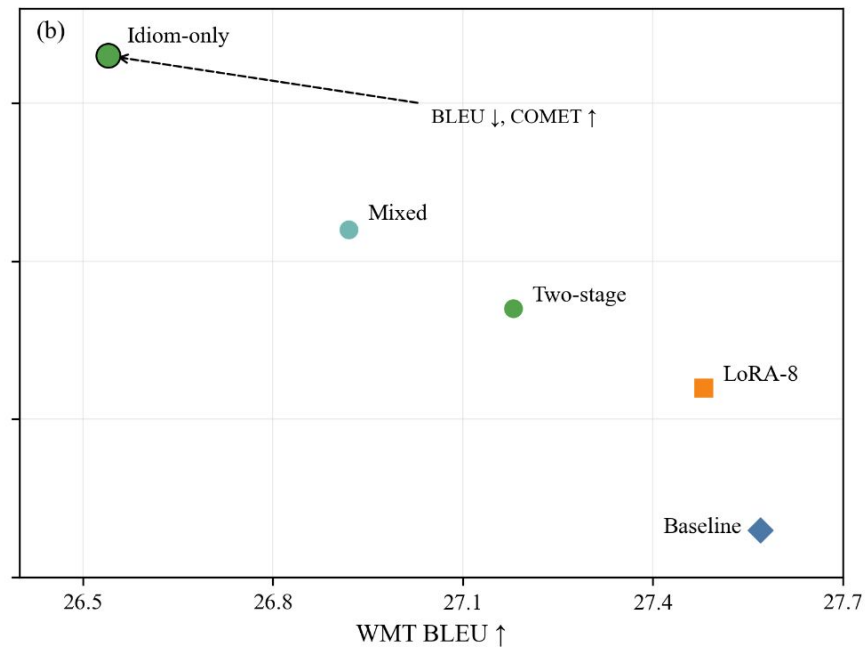
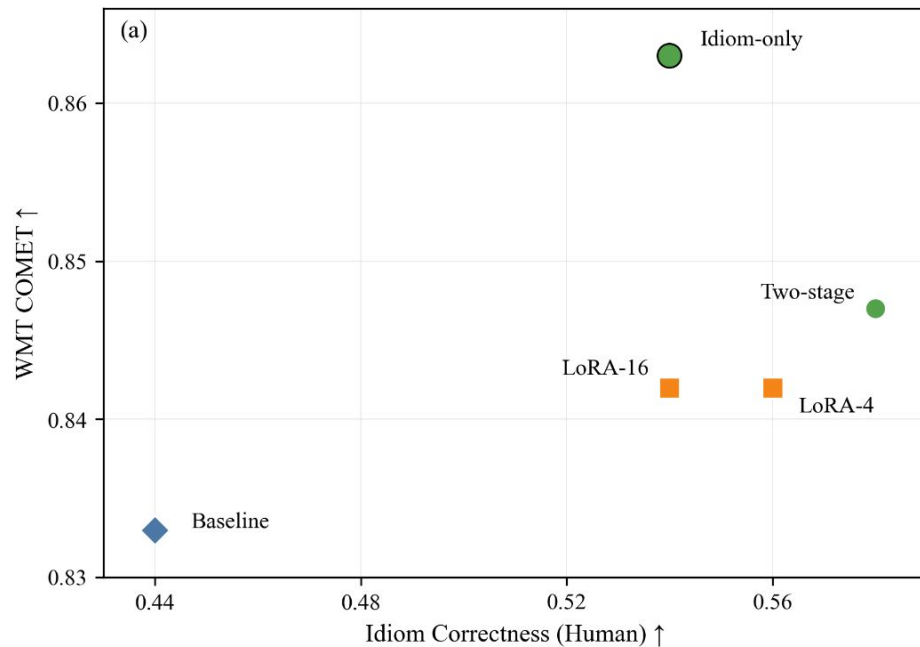
Human Evaluation

Model	Fluency	Adequacy	Idiom Correct.
Baseline	1.78	1.52	0.44
Idiom-only	1.82	1.54	0.54
LoRa-4	1.84	1.54	0.56
LoRA-8	1.80	1.58	0.58
LoRA-16	1.80	1.58	0.54
Prompt	0.68	0.44	0.02

Table 2: Human evaluation (25 idioms + 25 WMT).

- Two-stage fine-tuning achieves the highest adequacy and idiom correctness while maintaining fluency.

Results



Extra: other things we tried

Use a more powerful NMT model

- Facebook NLBB
- Got a critical system warning ("Core about to melt - do not proceed")
- Took too long to train in Colab

Test fine-tuned model on **English-Dutch sentence pairs**

- We thought that the we can apply the model gains to a language similar to German
- Experiment failed because of

Contribute a **corpus of 1.000-10.000 EN-DE idiom sentence pairs**

- We wanted to use the fine-tuned model to code a corpus builder and contribute a new dataset to the research community
- Idea failed, because the overall translation quality was not good enough
- Ran out of compute when trying to train more powerful base models (e.g. Facebook NLBB)

Thanks!

Backup

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Sentence Predictions

Example 1 (Idiom)

Source: Bend the knee, peasant! Admit that I am your rightful king!

Reference: Beuge das Knie, Bauer! Gib zu, dass ich dein rechtmäßiger König bin!

Baseline: Gib zu, dass ich dein rechtmäßiger König bin!

Two-stage: Kniebeugen, Bauer! Gib zu, dass ich dein rechtmäßiger König bin!

Idiom-only: Kniebeugen, Bauer! Gib zu, dass ich dein rechtmäßiger König bin!

LoRA r=16: Gib zu, dass ich dein rechtmäßiger König bin!

COMET: 0.799 / 0.917 / 0.917 / 0.799

This example highlights a different type of error. The baseline and LoRA outputs omit the opening imperative entirely, losing part of the source meaning. In contrast, the two-stage and idiom-only models retain more of the sentence structure, but render “bend the knee” as the literal and stylistically awkward “Kniebeugen.” Although these outputs are not fully natural, they better preserve the intent of the original sentence. This trade-off between completeness and idiomatic naturalness is reflected in the COMET scores.

Example 2 (WMT)

Source: “Decisions will then be made elsewhere,” said Maier.

Reference: “Dann werden Entscheidungen an anderer Stelle getroffen”, sagte Maier.

Baseline: “Beschlüsse werden dann an anderer Stelle getroffen werden”, sagte Maier.

Two-stage: “Die Entscheidungen werden dann an anderer Stelle getroffen werden”, sagte Maier.

Idiom-only: “Die Entscheidungen werden dann an anderer Stelle getroffen werden”, sagte Maier.

LoRA r=16: “Dann werden Entscheidungen anderswo getroffen”, sagte Maier.

COMET: 0.936 / 0.942 / 0.942 / 0.943

In this general-domain example, all models produce plausible translations, but differ in fluency and conciseness. The LoRA model yields a more natural formulation (“anderswo getroffen”), while the other outputs remain slightly redundant due to repeated auxiliary constructions. Although these differences are subtle, they affect readability and are reflected in the small but consistent differences in COMET scores.